



• Off-road autonomy chassis

LUCHS B

**BUILT FOR AUTONOMY.
FROM THE WHEELS UP.**

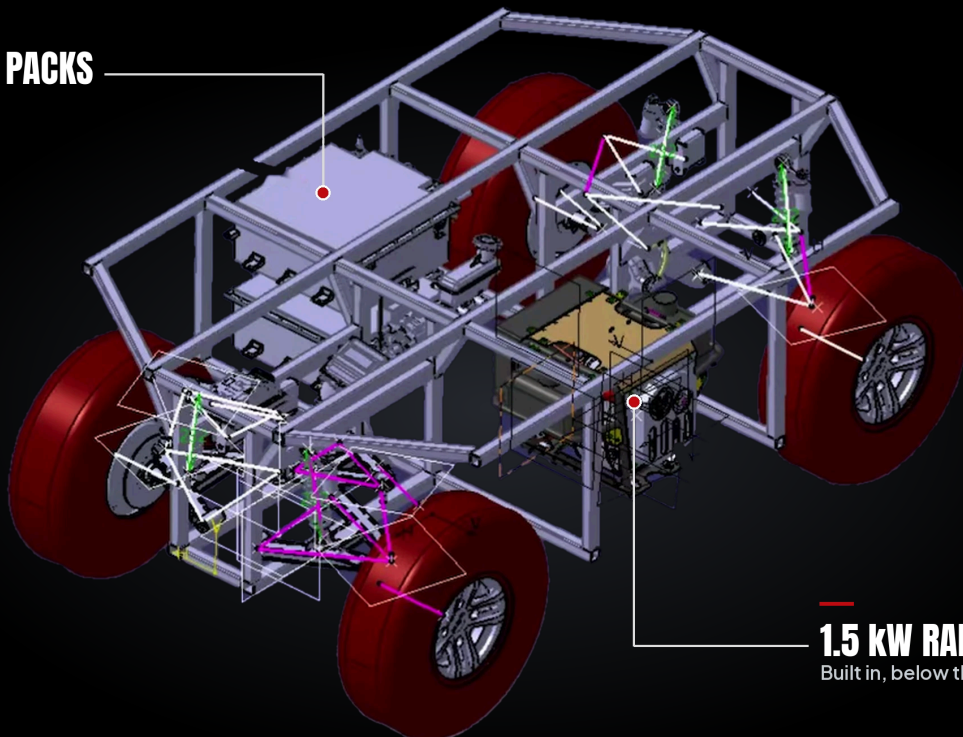
A full-size 4x4 drive-by-wire chassis: four in-wheel motors, steering on both axles, 200 kg maximum payload and one documented CAN interface shared across the Luchs family.

CHASSIS BY RJ TECH LIMITED / AUTONOMY BY TERRAAGENT ROBOTICS (TAR)

4x4

TWO BATTERY PACKS

About 5 kWh each



1.5 kW RANGE EXTENDER

Built in, below the deck

Desert mode in CAD: two battery packs and a 1.5 kW range extender, built into the chassis.

200 kg

Maximum payload; 80 kg working payload

40 km/h

Top speed, fully loaded

30 °

Maximum climb, fully loaded

300 mm

Ground clearance

300 km

Range in desert mode, with the built-in range extender; 60 km+ on battery

4* 330 N·m

Peak torque, one in-wheel motor per wheel

A FULL-SIZE OFF-ROAD BASE, BUILT FOR AUTONOMY FROM THE WHEELS UP.

Luchs B carries up to 200 kg across dirt, gravel and slopes, and takes its commands from your computer, not a driver. Four in-wheel motors, steering on both axes, an electro-hydraulic brake and an electric parking brake are all controlled through one documented interface. Mount your sensors and compute on the deck and spend your time on the autonomy, not the vehicle.

- **Steers like a car, turns tighter than one.** Two EPS units steer the front and rear axles. Choose single (front) or dual Ackermann in software.
- **Four in-wheel motors.** 1.8 kW rated and 5 kW peak at every wheel, 20 kW for the vehicle. 4WD or 2WD, selectable.
- **Goes the distance.** Battery for a working shift. Desert mode adds a second battery pack and a 1.5 kW range extender inside the chassis, for up to 300 km.
- **One interface for the whole family.** The Luchs Chassis Interface (LCI) runs over CAN with the same messages as Luchs A, so the same driver and the same code drive both. Available with the LCI firmware.



Luchs B on a dirt track: slopes, loose ground and a loaded run. Watch the film: Luchs_B_Promo_EN_1080p_v3.2_20261005.mp4.



Off the road. Loose dirt on a steep ramp, all four wheels driving. Range extender on the deck for testing.



CHOOSE HOW FAR IT NEEDS TO GO.

Every Luchs B shares the same frame, drivetrain, steering, brakes and interfaces. One battery pack or two: the number of packs sets the range. Desert mode adds a built-in range extender and goes furthest.

STANDARD

Luchs B Standard
One battery pack

Battery	LiFePO ₄ , 72 V 60 Ah
Energy	4.3 kWh
Payload power	12 V / 300 W
Range, full load	≥60 km

RECOMMENDED

Luchs B Pro
Two battery packs, longer range

Battery	LiFePO ₄ , 73 V 127.4 Ah
Energy	9.3 kWh
Payload power	12 V / 600 W
Range, full load	On request

ULTIMATE RANGE

Desert mode
Two batteries plus a built-in range extender

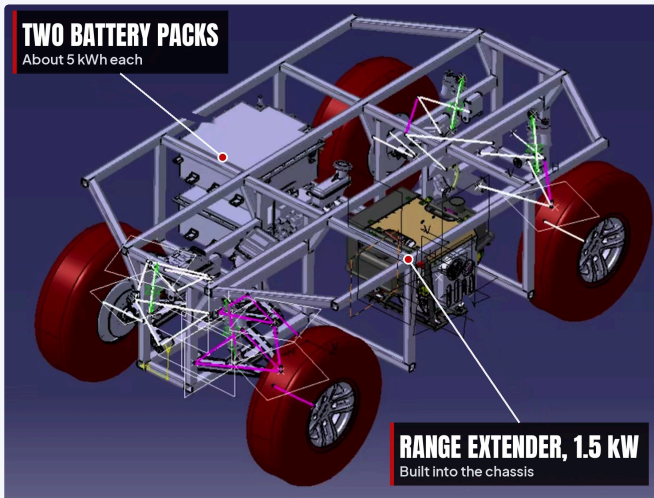
Battery	2 packs, about 5 kWh each
Range extender	1.5 kW, built in
Fuel	Petrol
Range	Up to 300 km

All configurations: 4 in-wheel motors, dual EPS, electro-hydraulic brake, electric parking brake, 80 kg working / 200 kg maximum payload. In desert mode, the range extender and fuel count against the payload.

• Desert mode

ULTIMATE RANGE. BUILT FOR THE DESERT.

Desert mode is the ultimate-endurance configuration: two battery packs of about 5 kWh each and a 1.5 kW range extender, all inside the frame. The range extender charges the batteries as you drive, so the vehicle keeps going far from a charger, and the deck stays free for your payload.



Desert mode in CAD. Two battery packs and the 1.5 kW range extender sit inside the frame.



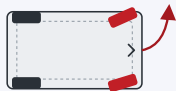
During testing the range extender is bolted to the deck so engineers can reach it. Production units carry it inside.

What you see in our photos and film: the black box on the deck is the range extender in its test position. On a delivered Luchs B it is out of sight, inside the chassis.

STEERING ON BOTH AXLES. DRIVE AT EVERY WHEEL.



Your planner chooses the motion model with a single command, and the reported odometry follows the model you selected.



Ackermann

Front axle steers, rear axle stays straight. Predictable, car-like arcs at speed.

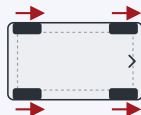
UP TO 40 KM/H



Dual Ackermann

Front and rear axles steer in opposite directions for a much tighter turn.

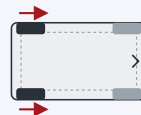
TURNING CIRCLE 4.4 M



4WD

All four in-wheel motors drive. Maximum traction for slopes, mud and loose ground.

DEFAULT OFF-ROAD



2WD

Two motors drive for efficient running on firm surfaces.



Skid-steer

Left and right wheels driven at different speeds to turn on the spot.

ON REQUEST

Creep mode gives steady very-low-speed travel on rough ground, slopes, mud, sand and rock.

BLACK OR WHITE. ONE CHASSIS.

A flat deck over a welded frame, independent suspension at every corner and 22-inch off-road tyres. Black is the standard version. White is the cooling version, with a heat-protective coating that keeps the battery and electronics cooler under a hot sun. A white unit with a black box on the deck is a test vehicle with its range extender exposed for debugging; every delivered unit carries it inside the chassis.



Black. Standard version.



Black, front.



White. Cooling version, test vehicle.



Rear three-quarter



Side

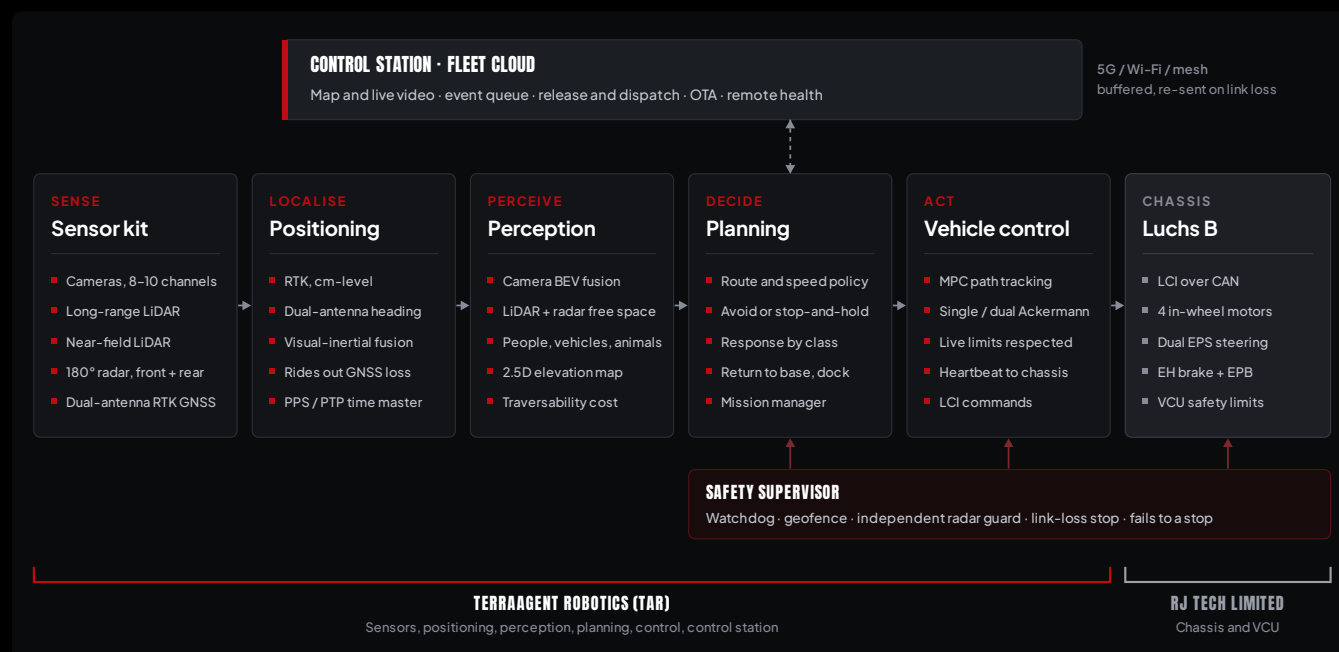


Front three-quarter

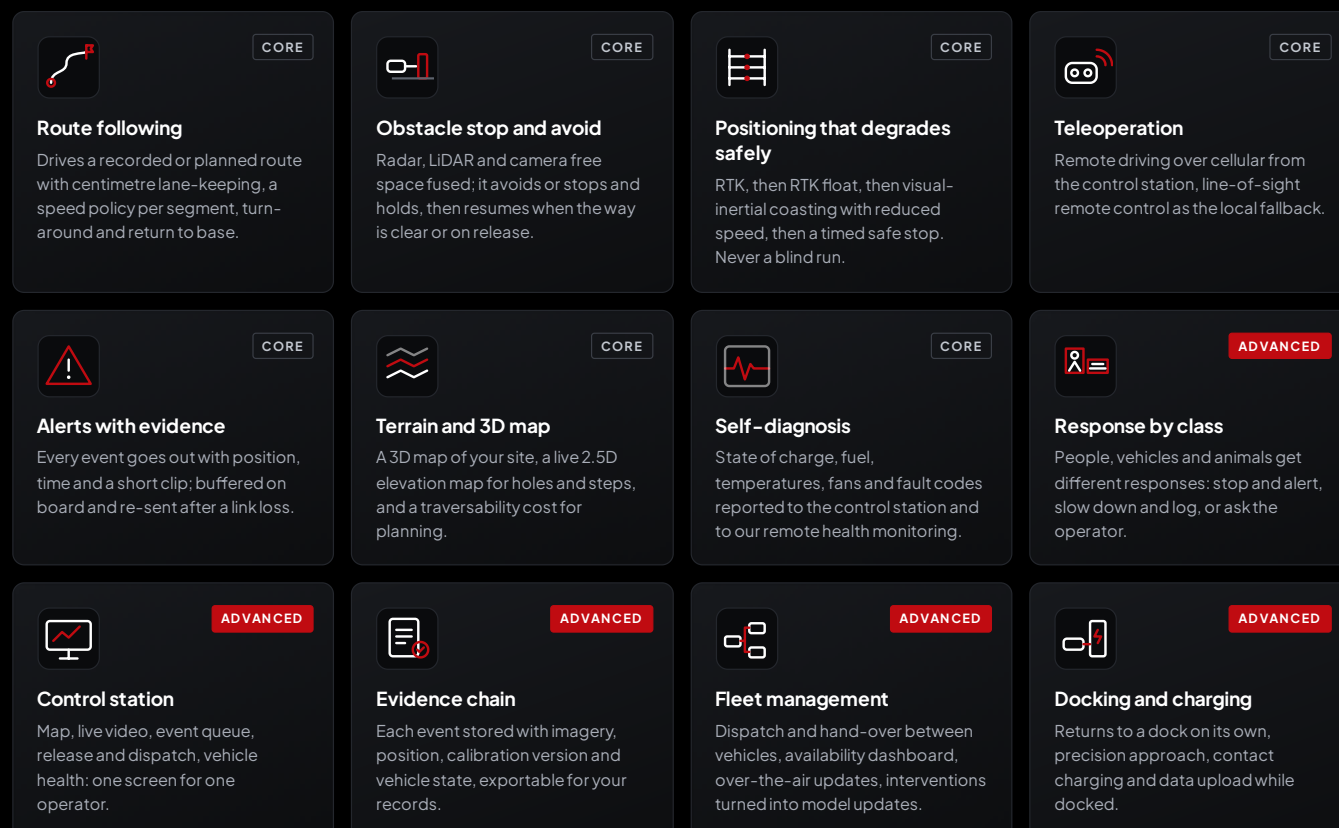
THE AUTONOMY IS OURS. BUILT BY TAR, ON LUCHS B.

RJ Tech builds the chassis. TerraAgent Robotics (TAR) builds what drives it: the sensor kit, positioning, perception, planning, vehicle control and the control station, integrated on Luchs B and delivered as one system. The building blocks already run on real vehicles, on real data.

01 THE STACK

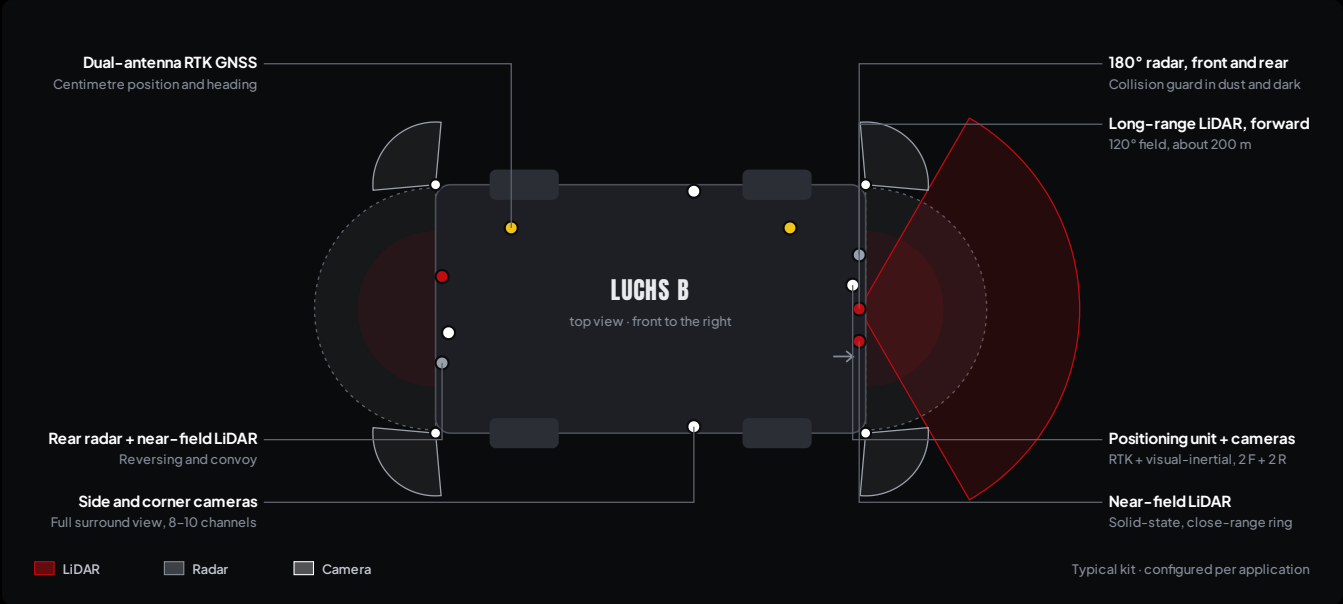


02 WHAT IT CAN DO



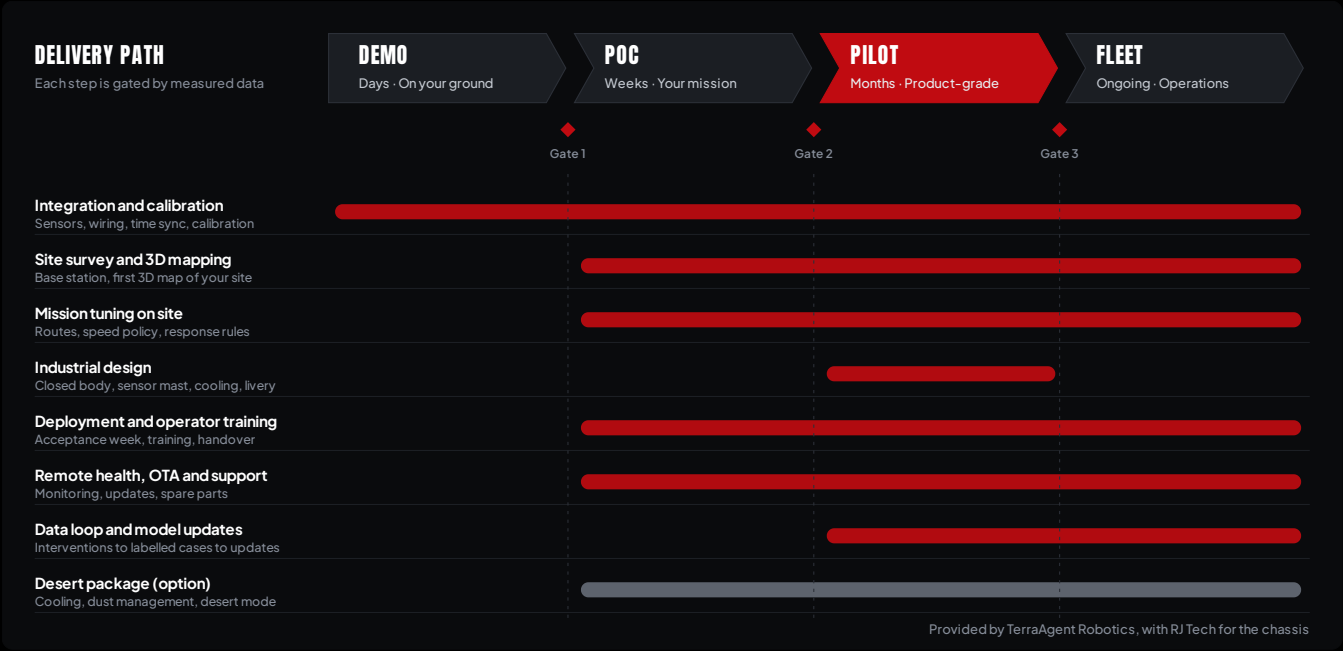
Core functions come with the TAR autonomy kit. Advanced functions are added for pilot and fleet operation.

03 THE SENSOR KIT

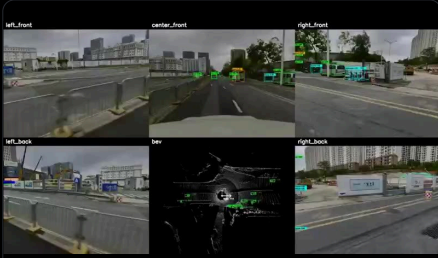


Optional mission kits: a dual-spectrum camera (visible and thermal) for detection at night and against low sun, and an acoustic camera that hears engines, generators or voices beyond the optical range. The sensor set gets simpler as it is proven: LiDAR concentrates in one lead vehicle that maps for the others.

04 BEYOND THE SOFTWARE



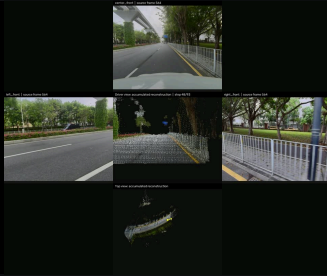
05 ALREADY RUNNING ON REAL VEHICLES AND REAL DATA



Camera-first perception

Five cameras and LiDAR fused into one bird's-eye view with 3D detection, projected back into every camera.

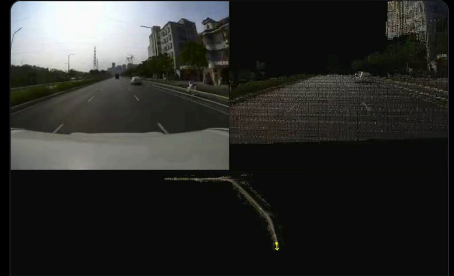
REAL VEHICLE



Dense 3D reconstruction

Three front cameras build a coloured 3D model of the road as the vehicle drives. No LiDAR.

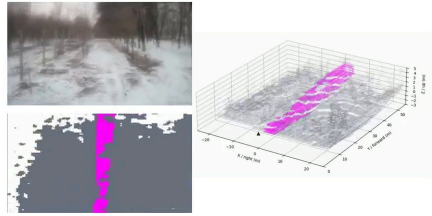
REAL VEHICLE



Reconstruction against low sun

The same reconstruction on a suburban road against the sun; the top view keeps the street shape through the turn.

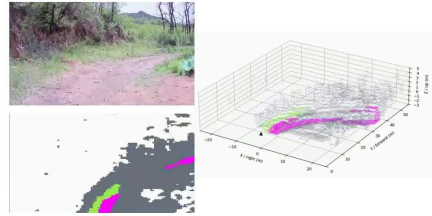
REAL VEHICLE



Drivable ground in snow

One camera, a 3D occupancy grid and a drivable-ground map (magenta) on a snow-covered forest track.

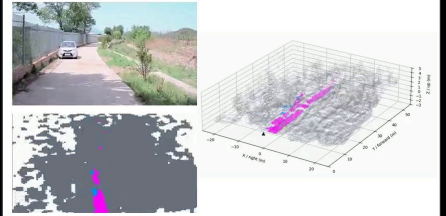
RECORDED OFF-ROAD DATA, PROCESSED OFFLINE



Dirt trail and verge

Follows a curving dirt track through the bends and tells the track from the verge (green).

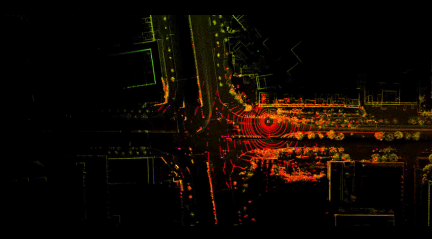
RECORDED OFF-ROAD DATA, PROCESSED OFFLINE



Oncoming vehicle

A narrow lane along a fence: the space taken by the oncoming car is cut out of the drivable ground.

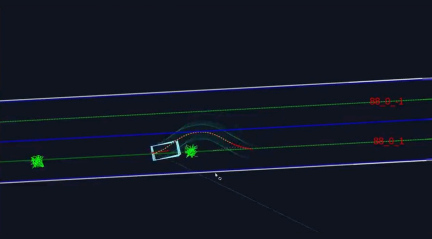
RECORDED OFF-ROAD DATA, PROCESSED OFFLINE



LiDAR map localisation

Each LiDAR scan registered to a prebuilt point-cloud map at 20 Hz: the second source of position when satellites drop out.

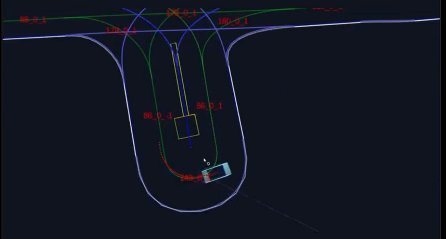
REAL VEHICLE, REAL TIME



Planning around an obstacle

The planner leaves a blocked lane, passes the obstacle and returns to the lane on a smooth path.

PLANNING STACK, REPLAY



Junctions, U-turns and parking

Route, behaviour and trajectory in three layers: turns at junctions, a U-turn and reversing into a bay.

PLANNING STACK, REPLAY

Real-vehicle tiles show our own software running on real vehicles. Off-road tiles are our perception model run offline on recorded off-road driving data. Planning tiles are replays of our planning stack. All are building blocks we integrate on the Luchs family.

• Field-proven

BUILT, TESTED, AND TAKEN TO THE DESERT BEFORE.

Luchs B comes from a team that builds custom chassis and converts vehicles for hard work. Our custom heavy and medium chassis are tested in the field, and our desert kit has already gone into vehicles we converted for hot, dusty work: sensor housings built into the bodywork, a sealed battery and electronics bay, and protected sensor windows.



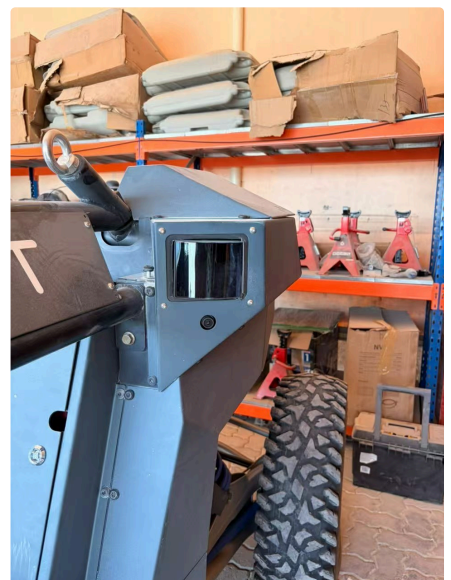
Custom chassis on field test. Heavy and medium platforms, mud and loose ground.



Desert kit: sensor housing built into the bodywork.



Desert kit: sealed bay for batteries and electronics.



Desert kit: protected sensor window.

• Workshop

BUILT IN OUR OWN WORKSHOP.

Frames, drive modules and battery packs are assembled and tested in our own workshop.



BOLT IT ON. PLUG IT IN. DRIVE IT FROM CODE.



A deck for your payload

A flat deck along the full length of the vehicle carries sensor masts, compute boxes and cargo.



Payload connector

Weatherproof connectors on the side panel for customer CAN and payload power. Planned: switched and fused 12 V and 24 V outputs plus a hard-wired external e-stop loop on one connector.



Commands at the motion level

Send a target speed with a yaw rate or road-wheel angles. The chassis closes the speed, steering and brake loops itself, enforces its own limits and tells you what they are.



Same code as Luchs A

LCI uses the same message IDs, layout and scaling across the Luchs family. A shared ROS 2 driver publishes odometry, battery and chassis state, and accepts velocity commands.

Interfaces at a glance

INTERFACE	DETAILS
Customer CAN	LCI v1.0: CAN 2.0B, 1 Mbit/s, big-endian, dedicated channel (today: 500 kbit/s, legacy messages)
Commands	Speed + yaw rate, or speed + front / rear road-wheel angles; gear; lamps and horn
Feedback	Speed and yaw rate, road-wheel angles, four wheel speeds, live limits, battery, faults, e-stop source
Heartbeat	Host frame every 20 ms; 200 ms silence → controlled stop and parking brake
Chassis IMU	Six-axis, 100 Hz on the customer CAN (option)
Payload power	24 V port today; 12 V and 24 V switched outputs planned
Remote control	2.4 GHz handheld transmitter included,

```
# Command 2.0 m/s forward, front-axle Ackermann (LCI 0x5A)
vx = 2000 # 0.001 m/s per bit → bytes 0-1
vy = 0 # unused on Luchs B → bytes 2-3
wz = 0 # 0.001 rad/s per bit → bytes 4-5
model = 0x02 # front Ackermann → byte 6
# → 07 00 00 00 00 00 02 00
```

LAYERED PROTECTION, BUILT IN.



The chassis watches itself and acts on its own: it limits what it is asked to do, stops when its host goes silent, and its emergency stops do not depend on any software.

Emergency stops everywhere

A button on the side panel, contact strips on the front and rear bumpers, a switch on the remote control and a host command. Any one brakes the vehicle hard and applies the parking brake.

Bumpers that think

After a front-bumper hit the vehicle can only reverse; after a rear-bumper hit it can only drive forward. You can always move away from what you hit.

Host heartbeat

If your computer stops sending its heartbeat for 200 ms, the chassis brings itself to a controlled stop and holds. Available with the LCI firmware.

Live limits

The chassis caps speed and steering for slope, temperature, battery and faults, executes commands at the limit rather than rejecting them, and reports the limits to your planner. Available with the LCI firmware.

Electric parking brake

Holds a fully loaded vehicle on a 20° slope for at least 5 minutes, facing up or down hill.

Self-test and diagnostics

Battery, motor controllers, steering and brakes are checked at power-on and monitored while driving. Faults are reported over the interface.

● Remote control

The handheld transmitter drives the vehicle. A switch hands control to your computer, and takes it back just as quickly.

● Maintenance

Motion locked out; used to clear faults and recover after an emergency stop. Planned. Today an emergency stop is cleared by switching the vehicle off and on.

● Automatic

Your computer drives over CAN. Status light (yellow / green / blue) planned.

TECHNICAL DATA

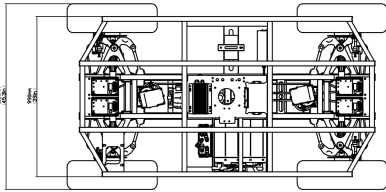


SIZE AND WEIGHT	
Dimensions (LxWxH)	1990 x 1150 x 800 mm
Wheelbase	1430 mm
Track width	970 mm
Ground clearance	300 mm
Tyres	22x7-10 in off-road
Chassis weight	230 kg
Working / maximum payload	80 kg / 200 kg
Suspension travel	50 mm, independent at each wheel
Versions	Black: standard · White: cooling version with heat-protective coating · Range extender built in (exposed on the deck only on test vehicles)
Approach / departure / ramp angle	90° / 90° / 40°
Maximum side tilt (full load)	30°
MOTION AND PERFORMANCE	
Drive	4WD, 2WD selectable · 4 in-wheel motors
Steering	Dual EPS · front Ackermann or dual Ackermann, selectable
Maximum road-wheel angle	30°
Steering resolution (wheel end)	0.1°
Minimum turning diameter	4.4 m (dual Ackermann)
Top speed (full load)	40 km/h
Maximum acceleration (no load)	0.8 g
Maximum climb (full load)	30° (58 %)
Maximum side slope while driving	40 % (22°)
Vertical step / trench / wading	150 mm / 400 mm / 300 mm
Service brake	Electro-hydraulic, 240 ms response
Parking brake	Electric parking brake; holds full load on 20° for ≥ 5 min
Creep mode	Yes
DRIVE AND POWER	
Drive motors	4 x in-wheel, 72 V · 1.8 kW rated / 5 kW peak each
Torque per wheel	130 N·m rated / 330 N·m peak
Peak system power	20 kW
Battery	LiFePO ₄ · 72 V 60 Ah (4.3 kWh) or 73 V 127.4 Ah (9.3 kWh) · >1000 cycles
Range	≥ 60 km (full load, 20 km/h, hard dirt) · ≥ 100 km (no load, 40 km/h, paved) · up to 300 km with range extender
Desert mode (option)	Two battery packs, about 5 kWh each, plus a 1.5 kW range extender built into the chassis · up to 300 km ·
Charging	On-board charger, full in 3 h · battery swap in under 3 min · AC inlet type per market
Payload power	12 V / 24 V switched outputs, fused
CONTROL AND COMMUNICATION	
Operating modes	Remote control · Automatic · Maintenance (planned)
Interface	LCI v1.0 over a dedicated CAN 2.0B channel, 1 Mbit/s, big-endian
Software	ROS 2 Humble / Jazzy and ROS 1 Noetic (luchs_ros 1.1: SDK, driver, robot model, simulator; one code base with Luchs A) · LCI and VCU protocol documentation · DBC files
Units before LCI firmware	VCU interface V2.6 over CAN, supported by the same software
Chassis IMU	Six-axis, 100 Hz
Remote control	2.4 GHz handheld transmitter, included
Lighting	Headlights, turn indicators, brake and reverse lights · stealth mode switches all lights off
SAFETY	
Emergency stop	Side-panel button · front and rear bumper strips · remote control · host command · external hard-wired loop (planned)
Protection	Power-on self-test and diagnostics · battery management · host heartbeat · live limits
ENVIRONMENT	
Operating temperature	-20 °C to +55 °C
Humidity	95 % (±3 %), tested at 20-30 °C
Ingress protection	IP67 below 440 mm (battery, motors, connectors) · IP65 for controllers
Maximum altitude	5000 m
Battery transport	UN38.3

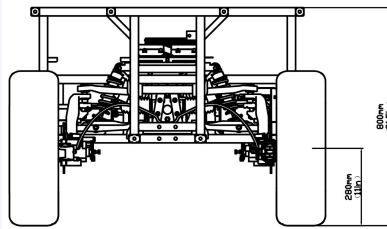
Typical values. Speed, range and gradeability depend on payload, surface, temperature and driving style. Specifications may change without notice. Other tyre sizes and steering configurations are available on request.

DRAWINGS

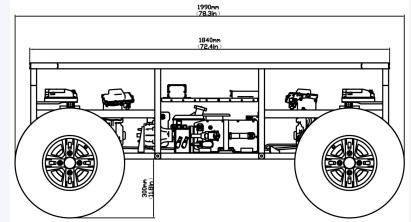
All dimensions in millimetres.



Top view



Front view



Side view

EVERYTHING YOU NEED TO START.



User manual

Safety, quick start, controls, safety functions, LCI and VCU interfaces, ROS software, TAR autonomy.

Luchs_B_User_Manual_EN_v0.9.5_20261005.pdf

Product brochure

This page as a printable PDF.

Luchs_B_Brochure_EN_v2.7_20261005.pdf

Data sheet

Three-page technical summary with the family line-up.

Luchs_B_DataSheet_EN_v1.4_20261005.pdf

CAN protocol (LCI)

All messages, signals and scaling, with a DBC file. Preliminary.

Luchs_B_CAN_Protocol_LCI_v1.0_draft.xlsx
x · .dbc

VCU interface V2.6

CAN interface of units shipped before the LCI firmware, with a DBC file.

Luchs_B_CAN_Protocol_VCU_V2.6.xlsx · .dbc

ROS2 software

SDK, driver, robot model, launch files, tools and documentation. Humble and Jazzy. One code base for Luchs A and Luchs B.

luchs_ros2_v1.1.0.zip

ROS1 software

The same SDK and driver for ROS1 Noetic.

luchs_ros1_v1.1.0.zip

Product film

About 40 seconds, 1080p and 720p.

Luchs_B_Promo_EN_1080p_v3.2_20261005.mp4

TERMS AND ABBREVIATIONS



2WD / 4WD Two- or four-wheel drive: how many wheels are driven.

Ackermann steering Car-style steering in which the inner wheel turns more sharply than the outer one, so both follow the same arc.

BMS Battery Management System: electronics that monitor and protect the battery.

CAN Controller Area Network: a robust two-wire bus widely used in vehicles and machines.

DBC A standard file format that describes CAN messages and signals for tools and software.

Dual Ackermann Both axes steer, in opposite directions, roughly halving the turning circle.

EPB Electric Parking Brake.

EPS Electric Power Steering: an electric motor that turns the steering.

Heartbeat A message the host sends at a fixed rate so the chassis knows it is alive.

IMU Inertial Measurement Unit: measures rotation rates and accelerations.

IP65 / IP67 Ingress Protection rating to IEC 60529. IP65: dust-tight and protected against water jets. IP67: dust-tight and protected against temporary immersion.

LCI Luchs Chassis Interface: the CAN interface shared by every Luchs chassis.

LiFePO₄ Lithium iron phosphate: a lithium-ion chemistry known for long cycle life and thermal stability.

Odometry An estimate of the vehicle's motion computed from wheel speeds and steering angles.

ROS 2 Robot Operating System 2: the most widely used open-source robotics middleware.

Yaw rate How fast the vehicle turns about its vertical axis, in rad/s.

• Contact

ONE FAMILY. TALK TO US.

Luchs B is the Luchs built for harder outdoor duty; Luchs A is the most versatile. One interface, one software stack. Tell us your application, and we will configure the chassis and the autonomy for it.

	Luchs A compact · most versatile	Luchs B this product	Luchs C custom
Built for	Robots that need many motion modes and interfaces, indoors and out	Off-road and site work: speed, range, clearance, heat and dust	Your specification
External dimensions	800×580×372 mm (A-4WS) 800×665×292 mm (other builds)	1990×1150×800 mm	Customisable
Chassis weight	40–55 kg	230 kg	Customisable
Working / max. payload	— / 80 kg	80 kg / 200 kg	Customisable
Max. climb grade	15°	30°	30°
Ground clearance	77 mm / 220 mm (A-4WS)	300 mm	Customisable
Steering	Skid-steer, Ackermann, dual Ackermann or four-wheel steer	Single or dual Ackermann (dual EPS)	Customisable
Max. speed	11.6 km/h	40 km/h	55 km/h
Drive	4 × 250 W hub motors	4 × in-wheel motors, 5 kW peak each	260 N·m×4 / 330 N·m×4
Battery	LiFePO ₄ , 48 V 15 Ah	LiFePO ₄ , 72 V 60 Ah / 73 V 127.4 Ah	LiFePO ₄ , up to 96 V 130 Ah
Endurance	Up to 6 h (no load)	≥60 km (full load), up to 300 km in desert mode	≥60 km, up to 300 km with range extender
Ingress protection	IP33 (IP54 build available)	IP67 (≤440 mm) / IP65 (>440 mm)	IP67
Host interface	CAN 1 Mbit/s (Luchs A protocol v1.1) · RS-485 / RS-232	CAN 1 Mbit/s, LCI v1.0 with heartbeat and live limits	LCI over CAN / CAN FD
Software	ROS 2 · ROS 1 · C++ SDK	ROS 2 · ROS 1 · C++ SDK	ROS 2 · ROS 1 · C++ SDK

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LUCHS B

Off-road autonomy chassis

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AUTONOMY

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